

PENGZHEN LI

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EDUCATION

University of Tennessee, Knoxville Ph.D. in Economics	expected 2027
Shandong University, Jinan M.A. in Finance — thesis published in <i>Technological Forecasting and Social Change</i> (2019)	2018
Tongji University, Shanghai B.A. in Economics	2015

RESEARCH INTERESTS

International Trade, Trade Policy, Applied Econometrics, and Political Economy.

WORKING PAPERS

Measured Quality Falls Within the Firm: Market Access and Chinese Exports with James Lake [**Job Market Paper**]. [[paper](#)]

Abstract: This paper links the 2002 removal of tariff uncertainty on Chinese exports to the United States to a decline in measured quality within continuing export relationships. Comparing a firm's US and non-US shipments of the same product, measured quality falls by about 6 percent among cells that ship in both periods in the three years before an overlapping quota removal, and by about 11 percent in the full sample over the whole window, which contains that removal. Restricting the sample to cells that ship in both periods retains 83 percent of the full-window estimate and yields a decline twice as large as the full-sample estimate in the three clean years. The price and quantity responses reject a markup cut on an unchanged demand schedule. Measured quality is a demand residual built from price and quantity, so it does not distinguish physical downgrading from a cheaper mix of varieties. We develop a model in which improved access can extend a seller's reach to buyers below a sorting boundary, who lower the return to appeal. Under stated conditions, appeal is capped and can fall. The continuing-cell price and quantity pattern is consistent with this mechanism but does not identify it against other sources of a demand-residual shift.

Trade Restrictiveness Indices with Discriminatory Tariffs with James Lake. *Presented at the Midwest Economic Theory and International Trade Meetings, Virginia Tech, 2025.* [[paper](#)]

Abstract: This paper extends the Trade Restrictiveness Index (TRI) to preferential tariffs. Using data for 2000–2020, a period of rapid expansion of Preferential Trade Agreements (PTAs), we compute 2,406 annual TRIs for 136 countries in an Armington model with large countries. Once terms-of-trade effects are taken into account, the global import-weighted TRI lies well below the average tariff, and the TRI for preferential tariffs declines faster than the TRI for MFN tariffs as the share of trade under PTAs grows. Across countries, average tariffs track the TRI poorly: the largest importers have TRIs far below their average tariffs, while many small economies have TRIs above them. Assuming that countries are small overstates the TRI, most of all for large importers.

A Mean-Field Game of Mayoral Endorsements: Identification and Evidence from Brazilian Coalitional Presidentialism with Matt Van Essen and Luiz Lima. *2024 J. Fred and Wilma Holly Award (best second-year paper).* [[paper](#)] [[slides](#)]

Abstract: A mayor's return from endorsing a presidential candidate falls as other mayors endorse the

same one, so endorsement is a strategic discrete-choice game that has gone structurally unestimated: among Brazil's 5,570 mayors the joint action space is too large to enumerate. Anonymity collapses the problem. When payoffs depend on rivals only through aggregate shares, equilibrium is a fixed point on the candidate simplex whose dimension does not grow with the number of players, and the model estimates by least squares in milliseconds. Estimating it on Brazil's 2014 election, we then show what the exercise does not buy: the prize parameter is algebraically the average log-odds of alignment rescaled by the share of mayors classified as aligned, so it changes sign with the classification rule. What survives is reduced form. A close-race difference-in-differences design puts discretionary transfers to co-partisan mayors 10.6 percent above those of other coalition members and finds no premium on the formula-based transfer that leaves no discretion to allocate.

PEER-REVIEWED PUBLICATIONS

[“Spatial optimization of the sustainable aviation fuel supply chains from forest residues via fast pyrolysis/hydrotreatment considering feedstock ash content variability.”](#) P. Li, T. E. Yu, N. Labbé, N. Abdoulmoumine, M. Garcia-Perez, K. P. Hoyt, and B. C. English. *Biomass and Bioenergy*, 208 (2026), 108793.

[“Assessing the impact of preprocessing and conversion technologies on the sustainable aviation fuel supply from forest residues in the Southeast USA.”](#) P. Li, T. E. Yu, C. Trejo-Pech, J. A. Larson, B. C. English, and D. N. Lanning. *Transportation Research Record*, 2678(9) (2024), 639–654.

[“Beef price spread relationship with processing capacity utilization.”](#) C. Martinez, P. Li, C. N. Boyer, T. E. Yu, and J. G. Maples. *Journal of the Agricultural and Applied Economics Association*, 2(1) (2023), 133–145.

[“Do green technology innovations contribute to carbon dioxide emission reduction? Empirical evidence from patent data.”](#) K. Du, P. Li, and Z. Yan. *Technological Forecasting and Social Change*, 146 (2019), 297–303.

PRESENTATIONS

Midwest Economic Theory and International Trade Meetings, Virginia Tech	Spring 2025
University of Tennessee, Economics Brown Bag Seminar Series	Spring 2024, Fall 2024
University of Tennessee, Applied Micro Group	Spring 2024, Fall 2024
Western Regional Science Association, Annual Meeting, Hawaii	2023
Western Agricultural Economics Association, Annual Meeting, Santa Fe, NM	2022
Southern Agricultural Economics Association, Annual Meeting, New Orleans, LA	2022

TEACHING EXPERIENCE

University of Tennessee

Instructor

ECON 351: Monetary Economics	Spring 2025, Spring 2026
ECON 322: The Global Economy: Trade and Development	Summer 2025
ECON 313: Intermediate Macroeconomics	Summer 2026

Teaching Assistant

Ph.D. courses:

ECON 511: Microeconomic Theory	Fall 2023, Fall 2024, Fall 2025
ECON 512: Advanced Microeconomics	Spring 2024
ECON 581: Mathematical Methods in Economics	Summer 2024

Master's courses:

PROFESSIONAL EXPERIENCE

University of Tennessee – Knoxville, TN

2021 – present

Graduate Research & Teaching Assistant. USDA NIFA and FAA ASCENT projects (2021–2023): mixed-integer supply-chain optimization (GAMS), IMPLAN/DEA siting analysis, and spatial data pipelines (ArcGIS, R, Python) for forest-residue sustainable aviation fuel; two publications and three working papers.

Shandong University – Jinan, China

2016 – 2018

Graduate Research Assistant. Panel-threshold and machine-learning analysis of green innovation and CO₂ emissions (published in *TFSC*, 2019).

HONORS AND AWARDS

Charles B. Garrison Award for Excellence in Teaching, University of Tennessee	2026
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Graduate Summer Fellowship, University of Tennessee	2025
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J. Fred and Wilma Holly Award (best second-year paper), University of Tennessee	2024
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J. Fred and Wilma Holly Fellowship, University of Tennessee	2023 – present
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Graduate Assistantship, University of Tennessee	2021 – present
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Outstanding Rainbow Volunteer, Tongji University (tutoring pupils in Shanghai)	2013 – 2015
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SKILLS

Programming Languages: Stata, R, Python, SQL, GAMS, MATLAB, LaTeX, ArcGIS, IMPLAN.

Languages: Chinese (native), English (fluent).

REFERENCES

James Lake (Chair)

Associate Professor of Economics

Department of Economics

University of Tennessee

Email: jlake7@utk.edu

Georg Schaur

Leigh Burch III Economics Professor and Haslam Family Faculty Fellow

Department of Economics

University of Tennessee

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Matthew Van Essen

Lena M. and Clarence A. Garvin, Sr. Professor and Dr. Joe Johnson Faculty Fellow

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